

SafeSign – Methodology

End-to-End Handwritten Signature Verification System

Our system is designed to verify the authenticity of handwritten signatures (specifically for high-stakes documents like bank cheques) by comparing a "query" signature against a "reference" signature of a known user. Our end goal is integrated the model with web application for real world deployment.

We are using **Siamese Neural Network** to find similarity between signatures. Unlike standard classifiers that categorize an image into a class, this network learns to calculate a high-dimensional feature vector (embedding) for two different input images. By measuring the distance between these two vectors, the system determines if the signatures belong to the same person (Genuine) or it is forged.

Dataset:

We are using combine dataset of GPDS150 and CEDAR.

Dataset	No. of Users	Samples per user
GPDS150	150	16 genuine, 16 forged
CEDAR	55	24 genuine, 24 genuine

Total images: 7440

We have also trained model on each dataset (GPDS150 and CEDAR) separately. We currently have more datasets like BHSig250 - Bengali and Hindi, we will use them for training later.

Working:

Before training, we analyzed the dataset characteristics to understand the variance in signature styles and image quality across **GPDS150** and **CEDAR** datasets.

- Resolution & Aspect Ratio Analysis: Calculating height/width distributions to inform the padding and resizing strategy.
- Pixel Intensity & Ink Density: Comparing the pixel histograms of genuine vs forged samples. Genuine signatures often exhibit more fluid ink density compared to the hesitations found in skilled forgeries.
- Distance Distribution: Establishing a baseline by calculating the initial distances between genuine-genuine pairs vs genuine-forged pairs using raw pixel values.

Data Preprocessing & Preparation

1. Grayscale Conversion: Removing color channels to focus on stroke patterns.
2. Binarization (Otsu's Thresholding): Isolating the signature from the paper background.

3. Inversion: Mapping ink to white (255) and background to black (0) for efficient sparse processing.
4. Aspect-Ratio Preserved Resizing: Using padding to scale images to a uniform size (224x224) without distorting the signature's unique slant or width.

The model requires pairs of images to learn similarity, so we create Siamese pairs:

Positive Pairs (Label 1): genuine – genuine (same user).

Negative Pairs (Label 0): genuine – forged (same user).

Model Architecture Selection

We have used three separate base CNNs for Siamese network:

1. **Custom CNN**: lightweight model optimized for faster inference.
2. **ResNet-18**: prevents vanishing gradients, suitable for deep feature extraction.
3. **EfficientNet-B0**: balances depth, width and resolution for better accuracy.

The twin networks share identical weights. When two images are passed through, the model calculates embeddings for both images.

The similarity is quantified through Euclidean Distance. If the Euclidean distance between two embeddings is below the threshold, the signature is classified as Genuine; otherwise, it is classified as Forged. We have set threshold to 0.5 currently.

Individual Contribution:

Sumanta Rakshit: Data Collection, Data Preprocessing, Modelling, Literature Survey, Report.

Aaditya Kumar Choudhary: Data Collection, Literature Survey, Building UI for initial implementation, Presentation(ppt).

Sachin Kumar: EDA, Data Preprocessing, Modelling, Literature Survey, Report.

Sayandeep Biswas: EDA, Website for initial implementation, Literature Survey, Presentation(ppt).